

MULTI-ECHELON SUPPLY CHAIN NETWORK DESIGN

Business Analytics – WEEK-3 Task | Mixed-Integer Linear Programming (MILP)

1. Introduction

Multi-Echelon Supply Chain Network Design is an optimization problem in which products move through multiple stages such as factories, warehouses, and customer zones. The objective is to determine the best shipment quantities and facility decisions while satisfying production, transportation, storage, and customer service requirements. Mixed-Integer Linear Programming (MILP) is suitable because it can combine continuous shipment quantities with binary decisions such as whether a warehouse is opened.

2. Problem Statement

Design a global logistics network containing factories, warehouses, and customer zones. The model must represent multi-echelon transport limits, warehouse storage capacities, and minimum service-level requirements. The optimization model minimizes total supply-chain operating expenditure while meeting all business constraints.

3. Objectives

- Minimize total production, transportation, and warehouse operating costs.
- Determine optimal product flows from factories to warehouses and from warehouses to customers.
- Respect factory production and transportation capacity limits.
- Respect warehouse storage capacities.
- Maintain minimum customer service-level requirements.
- Demonstrate the use of MILP for supply-chain network optimization.

4. Network Structure

Echelon	Entity	Role
1	Factories	Produce products and supply warehouses
2	Warehouses	Receive, store, and distribute products
3	Customer Zones	Receive final products and require minimum service levels

5. Decision Variables

Let x_{fw} = quantity transported from factory f to warehouse w .

Let y_{wc} = quantity transported from warehouse w to customer zone c .

Let $z_w \in \{0,1\}$ = 1 if warehouse w is opened/used; otherwise 0.

6. Objective Function

$$\text{Minimize } Z = \sum_{f,w} C_{fw} x_{fw} + \sum_{w,c} C_{wc} y_{wc} + \sum_w F_w z_w + \text{production costs}$$

Here, C_{fw} and C_{wc} represent transportation costs, F_w represents fixed warehouse operating cost, and production costs represent factory operating expenditure.

7. MILP Constraints

Constraint	Mathematical Representation
Factory capacity	$\sum_w x_{fw} \leq \text{Supply}_f \quad \forall f$
Warehouse flow balance	$\sum_f x_{fw} = \sum_c y_{wc} \quad \forall w$
Warehouse capacity	$\sum_c y_{wc} \leq \text{Capacity}_w \cdot z_w \quad \forall w$

Customer demand/service	$\sum_w y_{wc} \geq \text{ServiceLevel}_c \cdot \text{Demand}_c \quad \forall c$
Transport limit	$x_{fw} \leq \text{TransportLimit}_{fw} \quad \forall f, w$
Non-negativity	$x_{fw} \geq 0, y_{wc} \geq 0$
Binary decision	$z_w \in \{0,1\}$

8. Constraint Matrix Representation

The MILP can be written in standard matrix form as:

Minimize: $c^T x + f^T z$
Subject to: $Ax + Bz \leq b$
 $Ex + Gz = d$
 $x \geq 0$
 $z \in \{0,1\}$

The matrix A contains coefficients for shipment-related constraints, B contains coefficients associated with binary warehouse decisions, and b contains capacity and service bounds. Equality matrix E represents flow-balance relationships.

9. Sample Input Data

Factory	Capacity	Warehouse	Capacity	Customer	Demand
F1	500	W1	450	C1	250
F2	400	W2	350	C2	300
F3	300	W3	300	C3	200

Illustrative minimum service level: C1 = 90%, C2 = 95%, C3 = 90%. Transportation costs are assigned to each factory–warehouse and warehouse–customer connection in the optimization model.

10. Python MILP Implementation

```
import pulp

model = pulp.LpProblem("Supply_Chain_MILP", pulp.LpMinimize)

# Shipment variables
x = pulp.LpVariable.dicts("Factory_Warehouse", FW, lowBound=0)
y = pulp.LpVariable.dicts("Warehouse_Customer", WC, lowBound=0)

# Warehouse binary variables
z = pulp.LpVariable.dicts("Open_Warehouse", W, cat="Binary")

# Objective
model += (
    pulp.lpSum(cost_fw[f,w] * x[f,w] for f,w in FW) +
    pulp.lpSum(cost_wc[w,c] * y[w,c] for w,c in WC) +
    pulp.lpSum(fixed_cost[w] * z[w] for w in W)
)

# Factory capacity
for f in F:
    model += pulp.lpSum(x[f,w] for w in W) <= supply[f]

# Warehouse flow balance and capacity
for w in W:
    model += pulp.lpSum(x[f,w] for f in F) == pulp.lpSum(y[w,c] for c in C)
    model += pulp.lpSum(y[w,c] for c in C) <= warehouse_capacity[w] * z[w]

# Minimum service level
for c in C:
    model += pulp.lpSum(y[w,c] for w in W) >= service[c] * demand[c]

model.solve()
print(pulp.LpStatus[model.status])
print("Minimum Cost =", pulp.value(model.objective))
```

11. Optimization Process

- Define factories, warehouses, and customer zones.
- Enter production capacities, warehouse capacities, demands, service levels, and transportation costs.
- Create continuous shipment variables and binary warehouse variables.
- Construct the objective function to minimize total operating expenditure.
- Add capacity, flow-balance, service-level, and binary constraints.
- Solve the MILP using a solver such as CBC through PuLP.
- Analyze the optimal shipment plan and total minimum cost.

12. Expected Output

Output	Description
Optimal warehouse decisions	Warehouses selected for operation
Factory → warehouse flow	Optimal quantities shipped to warehouses
Warehouse → customer flow	Optimal quantities delivered to customers
Service level	Percentage of demand satisfied in each zone
Minimum total cost	Lowest feasible supply-chain operating expenditure

13. Business Analytics Interpretation

The optimized network helps management decide where inventory should flow and which warehouses should be utilized. The model balances cost efficiency with operational constraints. By changing demand, transportation costs, capacities, or service-level targets, analysts can perform scenario analysis and evaluate alternative supply-chain strategies.

14. Advantages

- Reduces unnecessary transportation and facility costs.
- Improves utilization of factories and warehouses.
- Ensures minimum customer service requirements.
- Supports data-driven network design decisions.
- Allows scenario and sensitivity analysis.

15. Conclusion

The Multi-Echelon Supply Chain Network Design problem can be effectively formulated as a Mixed-Integer Linear Programming model. The proposed model integrates factory capacities, warehouse storage limits, transportation constraints, and minimum service levels. Solving the MILP produces a cost-efficient logistics network and provides a practical Business Analytics framework for strategic supply-chain decision making.